



Project HAMi

从本地设备到远程算力池：远程GPU方案与HAMi 2.10新能力

李孟轩

密瓜智能联合创始人兼CTO & HAMi Maintainer

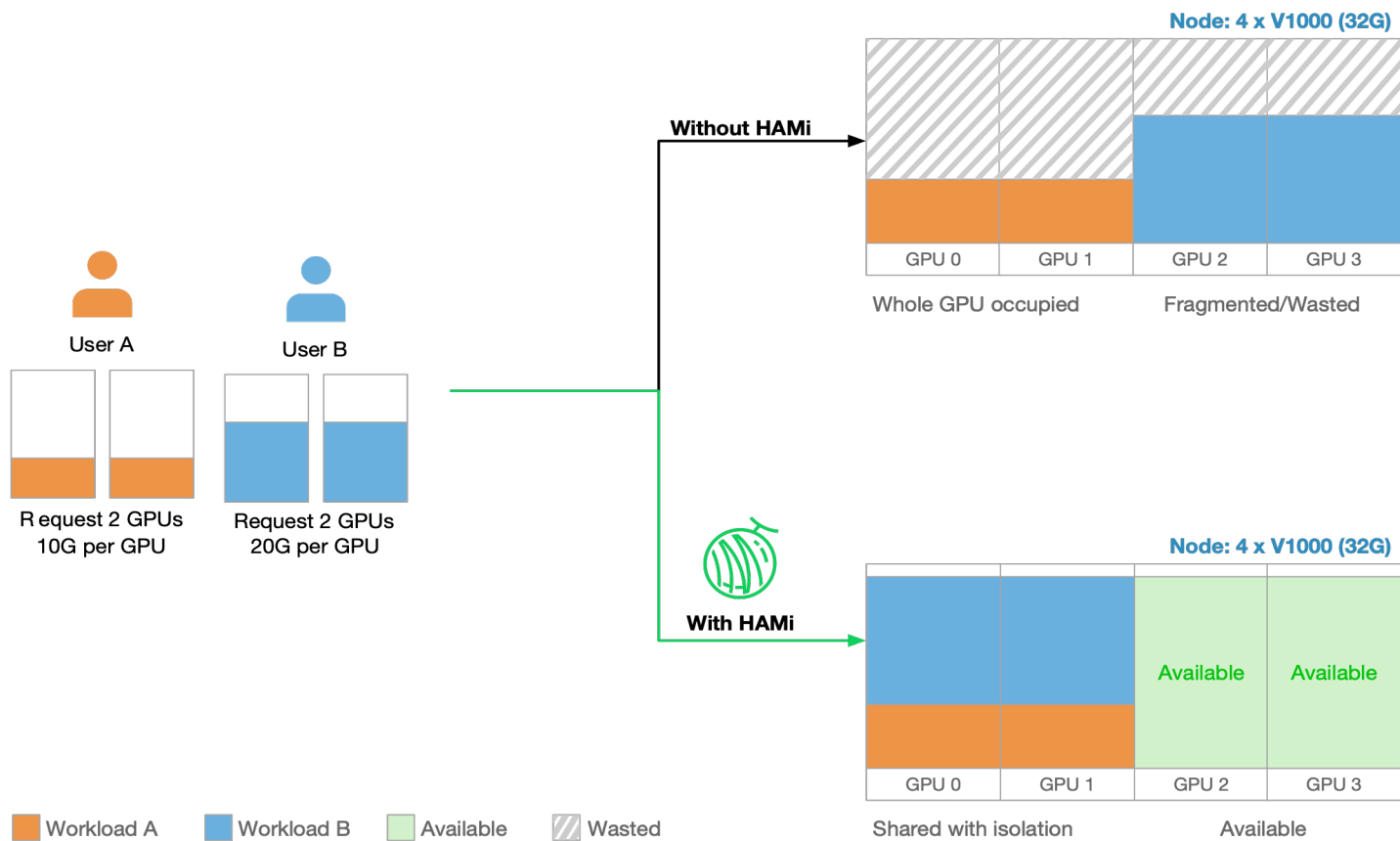
目录



-  HAMi介绍
-  HAMi v2.10新功能
-  DRA生态进展
-  HAMi v2.11 RoadMap

HAMi > Kernel

Kernel为做GPU时代的操作系统



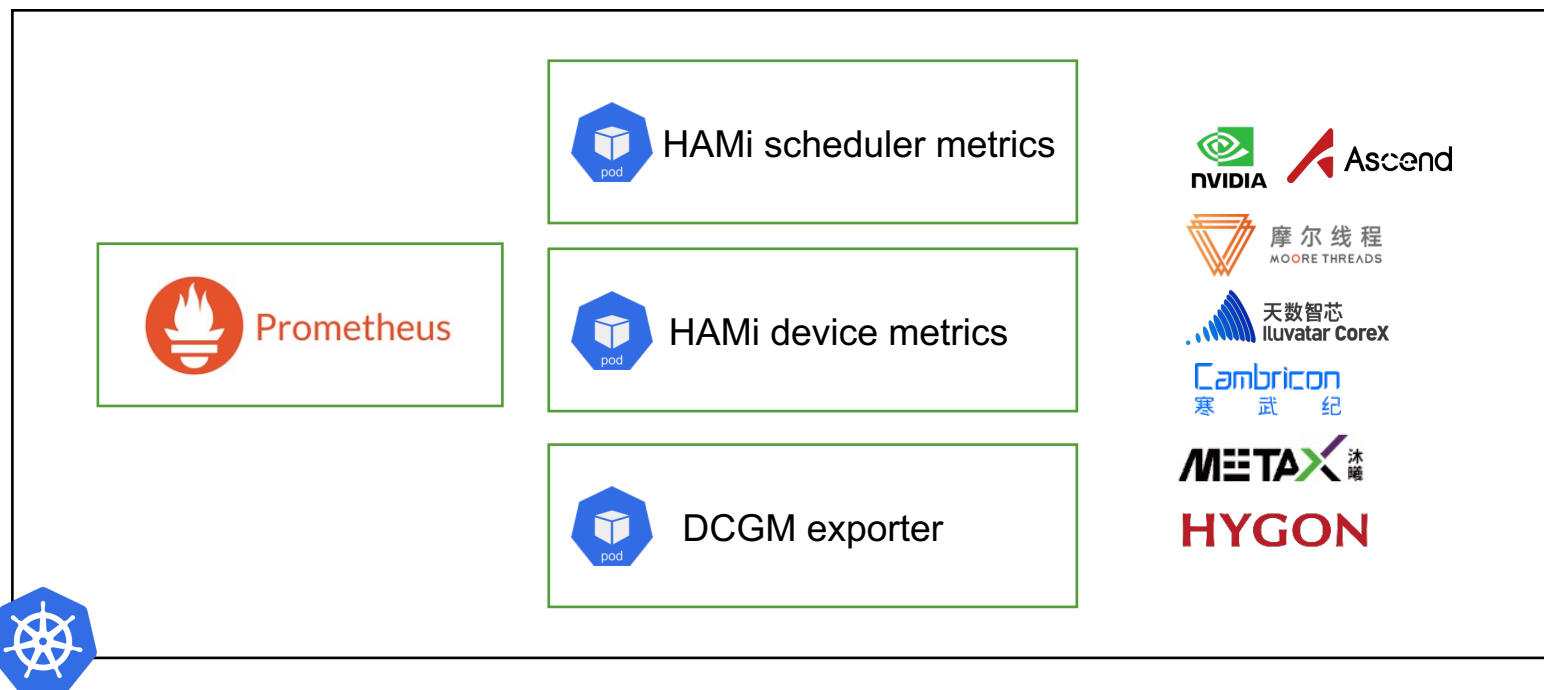
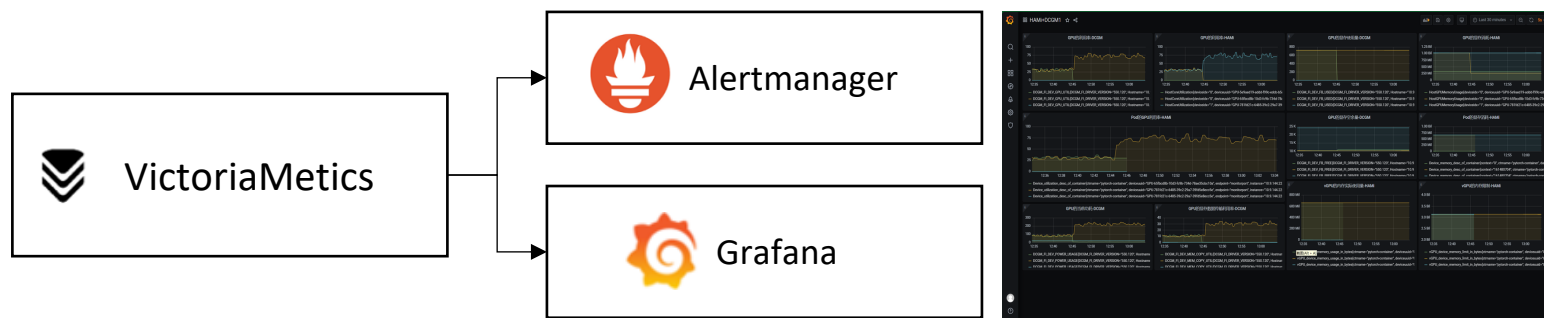
GPUS 需求4 → 2

利用率45% → 90%



HAMi可观测性 > 调度和分配

真实使用尽在掌握



01

K8s 调度维度: vGPU 都被哪些任务使用了, 任务、GPU 绑定关系

02

节点有三张卡, 每张卡被哪些任务占用、占用了多少算力、显存

03

GPU 设备维度: 反映应用运行过程中真实算力、显存使用率

04

A100 有三个任务, 每个任务真实使用了 20% 算力(时间片)、30% 显存

05

HAMi 也提供社区最佳实践 监控面板, 一站式观测调度与真实使用情况

各行各业的大模型应用 (400+)

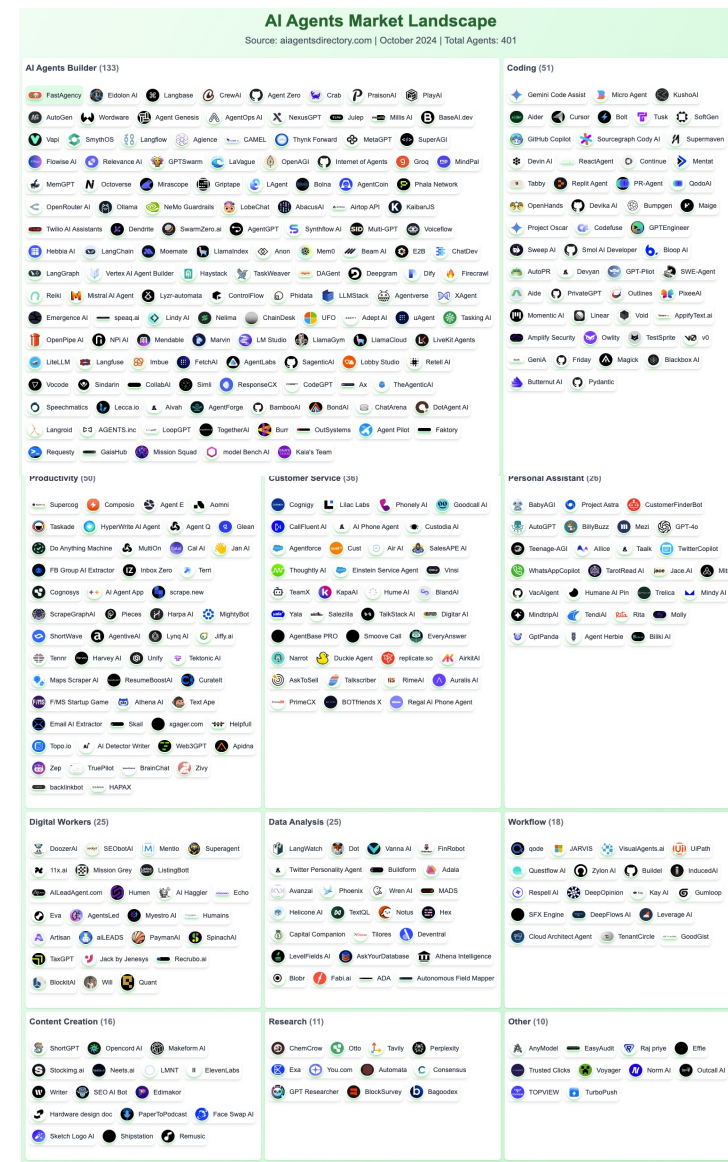
The diagram illustrates the AI ecosystem, categorized into three main sections:

- Model Providers (Left):** Includes logos for 豆包 (Dou Bao), Dify.AI, 百度文心一言 (Baidu Wenxin Yiyang), 智谱·AI (Zhipu AI), 百川智能 (Baichuan AI), 阿里云 Qwen (Alibaba Cloud Qwen), 百度文心一言 (Baidu Wenxin Yiyang), 智谱·AI (Zhipu AI), 百川智能 (Baichuan AI), 阿里云 Qwen (Alibaba Cloud Qwen), 无问芯穹 (Wu Wen Xin Qiong), Colossal AI, and Pytorch.
- Platform/Model (Middle):** Includes logos for 昇思 (MindSpore), 昇思大模型平台 (MindSpore Large Model Platform), 华为云盘古大模型 (Huawei Cloud Pangu Large Model), and 晟思 MindAIe (Shengsi MindAIe).
- APIs/Inference (Right):** Includes logos for Industry Standard APIs, Nvidia Models, Nvidia Models, Triton Inference Server, TensorRT LLM, and Pytorch.

The diagram illustrates the inference stack components, separated by a dashed line. On the left side (client-side), the components are INFINIGENCE 无问芯穹, Colossal AI, and Pytorch. On the right side (server-side), the components are 晟思 MindAI, Triton Inference Server, TensorRT LLM, and Pytorch.




Accelerator	Interface
Iluvatar CoreX (天数智芯)	CANN
METAX (沐曦)	CANN
Moore Threads (摩尔线程)	CANN
Cambricon (寒武纪)	CANN
Ascend	CANN
CUDA	CUDA
Nvidia GPU	CUDA



A person with a backpack is walking away on a path through a dense bamboo forest. The path is narrow and leads into the distance, flanked by tall, slender bamboo stalks. The lighting is soft, suggesting a misty or overcast day.

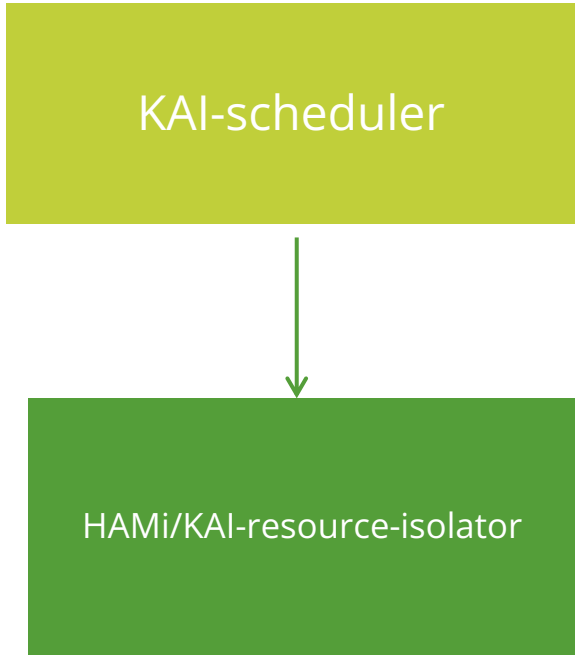
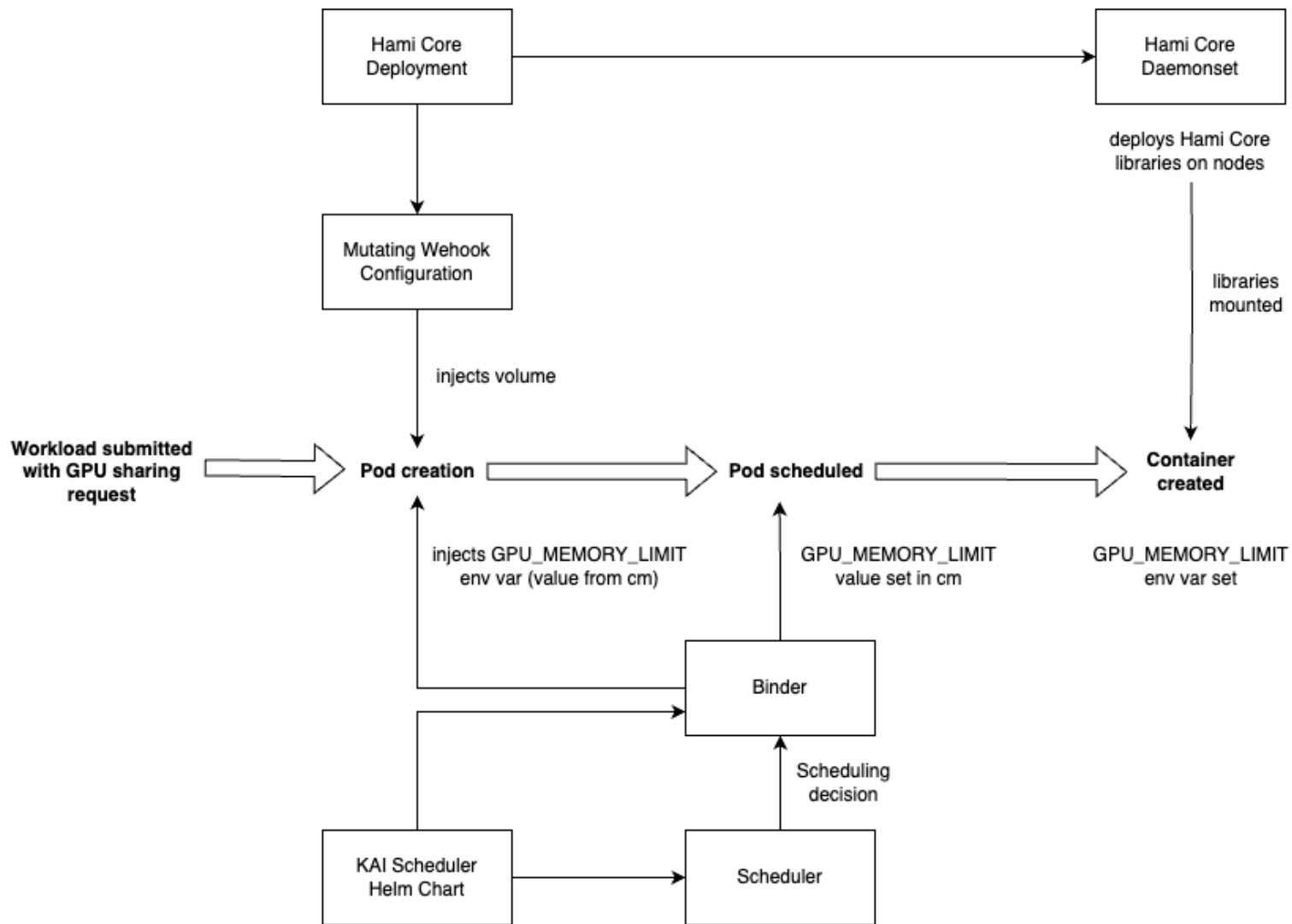
HAMi的新版本

HAMi v2.10

新特性

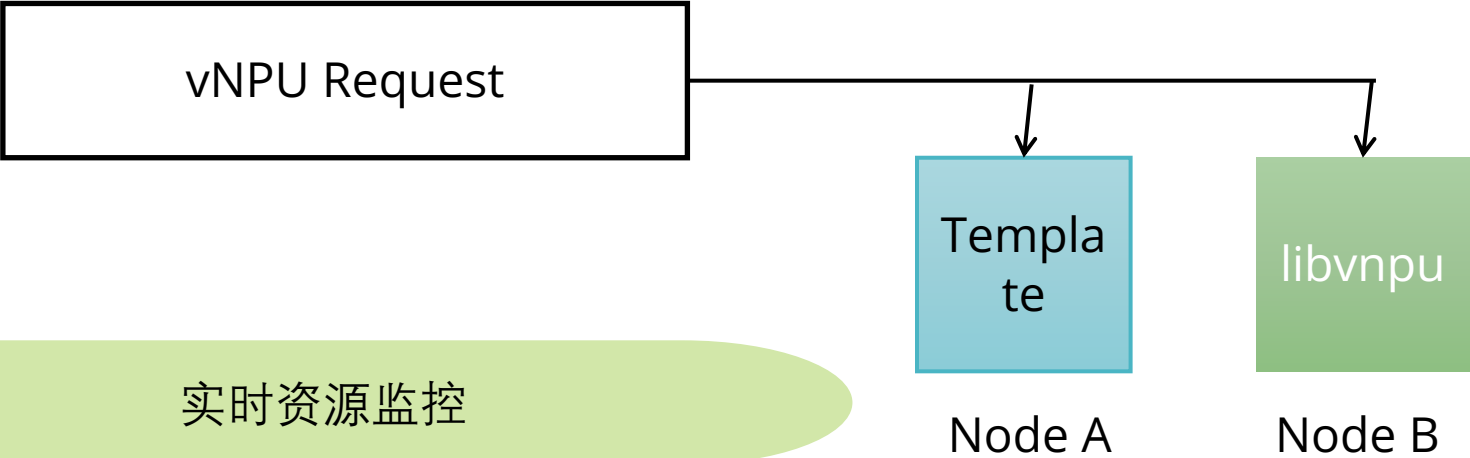
KAI-scheduler适配

Design: <https://github.com/kai-scheduler/KAI-Scheduler/pull/60>



vNPU 切分方案 全面升级

支持异构任务



实时资源监控

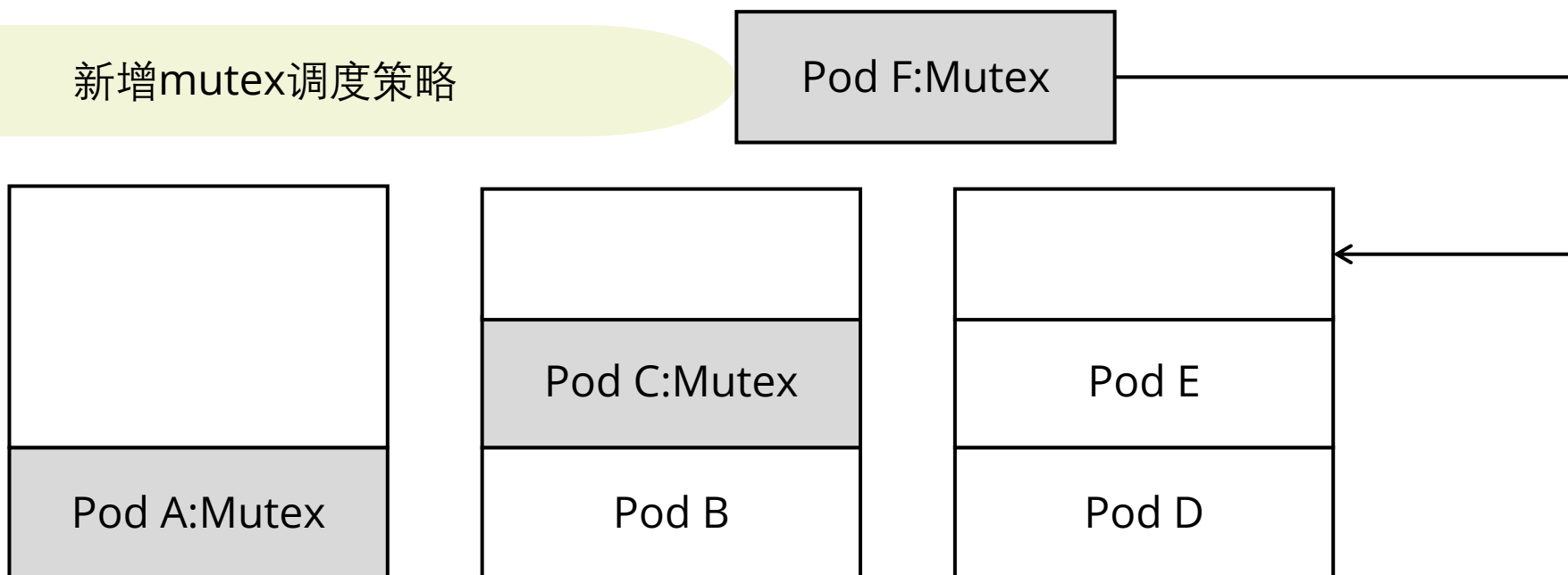
curl {cluster pod ip}:9395/metrics



Metric	Description
hami_host_gpu_memory_used_bytes	Physical NPU memory used (bytes)
hami_host_gpu_utilization_ratio	Physical NPU AICore utilization (0-100)
hami_vgpu_memory_used_bytes	Per-container vNPU memory used (bytes)
hami_vgpu_memory_limit_bytes	Per-container vNPU memory limit (bytes)
hami_container_device_utilization_ratio	AICore utilization of the device the container runs on (0-100)

调度全面升级

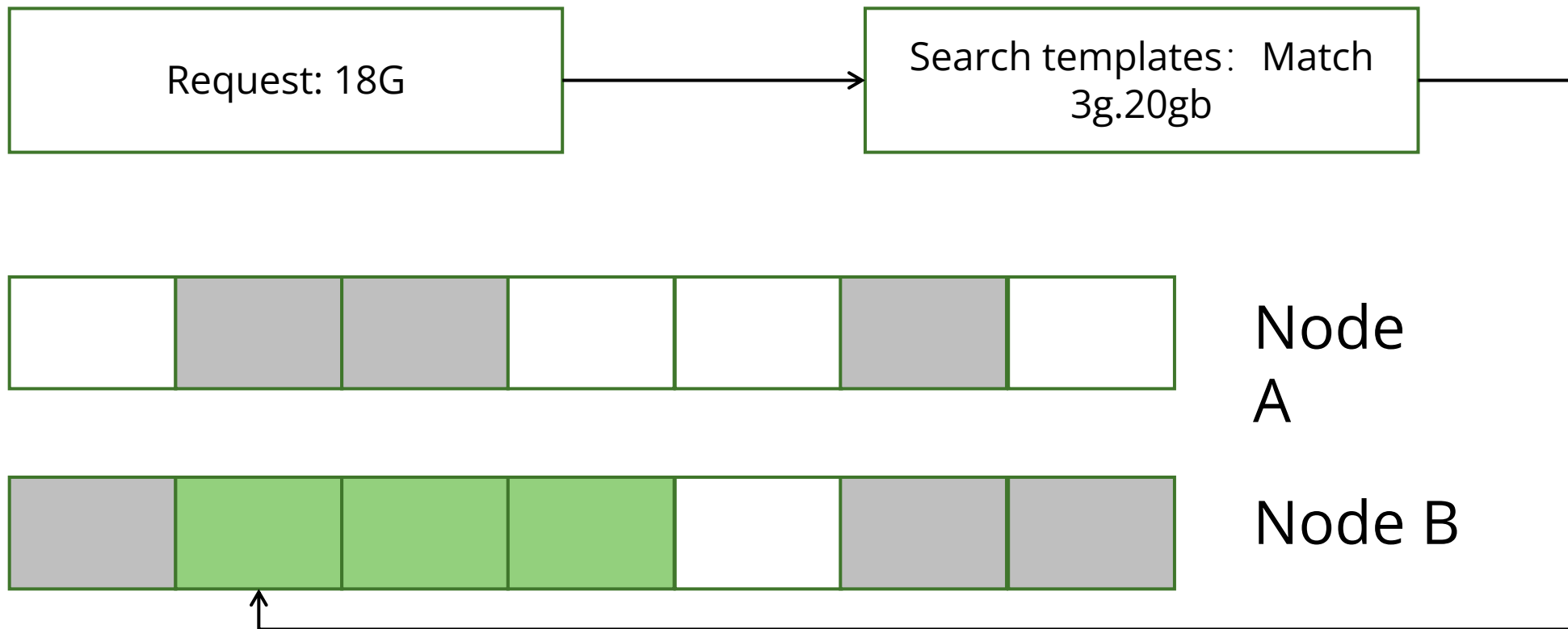
新增mutex调度策略



支持调度策略组合

`hami.io/node-scheduler-policy="spread,numa,mutex"`

更灵活的动态MIG

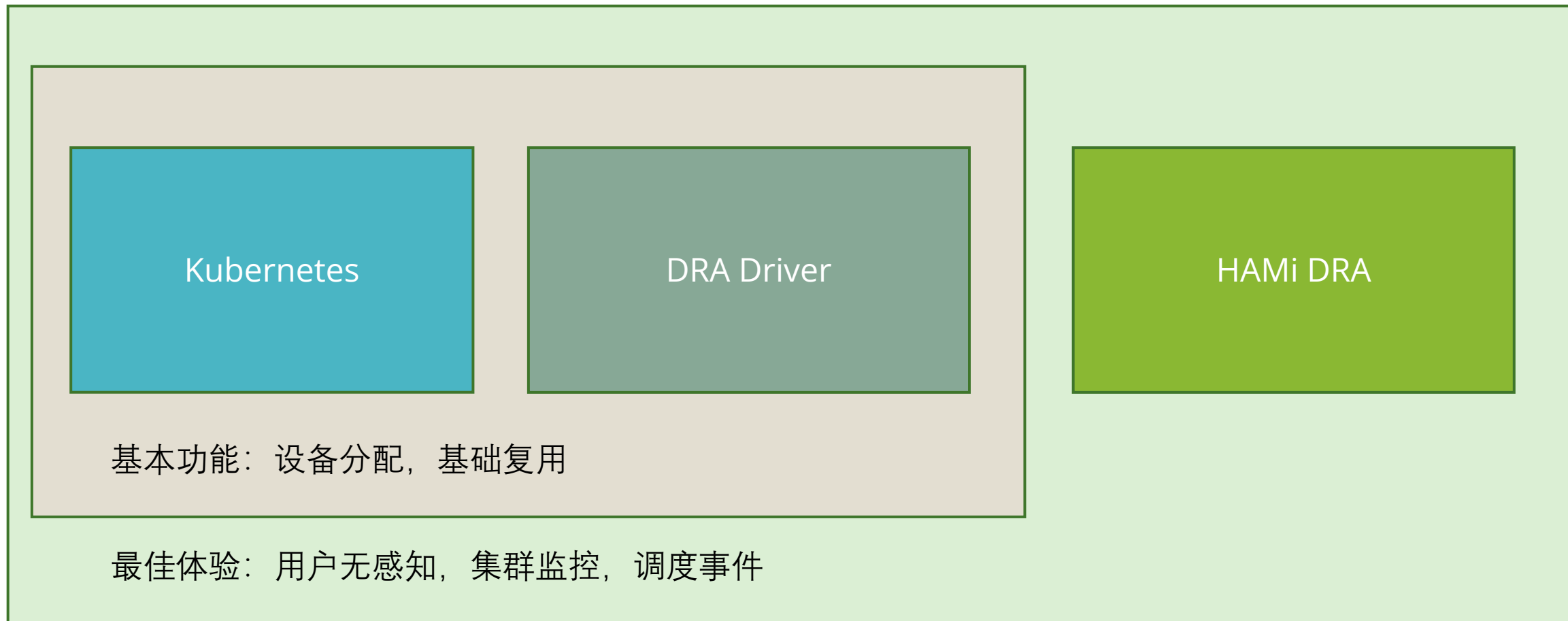


A person with a backpack is walking away on a path through a dense bamboo forest. The path is narrow and leads into the distance, flanked by tall, slender bamboo stalks. The lighting is soft, suggesting a misty or overcast day.

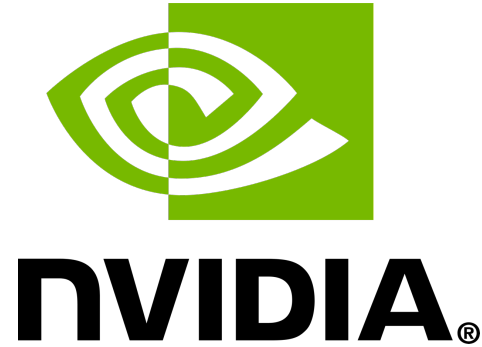
HAMi-DRA

DRA 生态进展

HAMi DRA解决方案



HAMi DRA: Ascend支持



HAMi DRA: InComing



HYGON

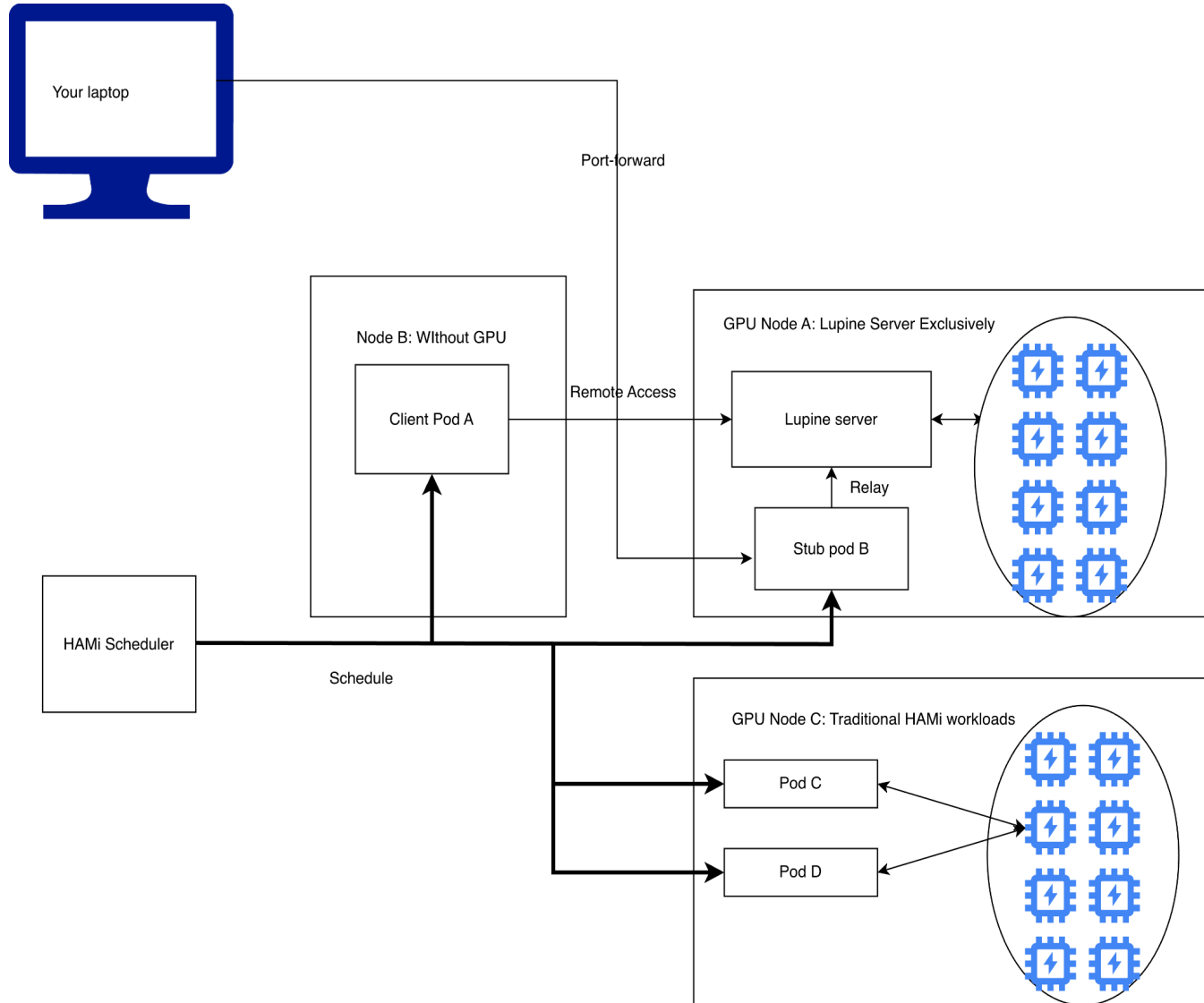


A person with a backpack is walking away on a dirt path through a dense bamboo forest. The path is narrow and leads into the distance, flanked by tall, slender bamboo stalks. The lighting is soft, suggesting a misty or overcast day.

HAMi RoadMap

v2.11 Roadmap

HAMi: Remote GPU



Port-forward:

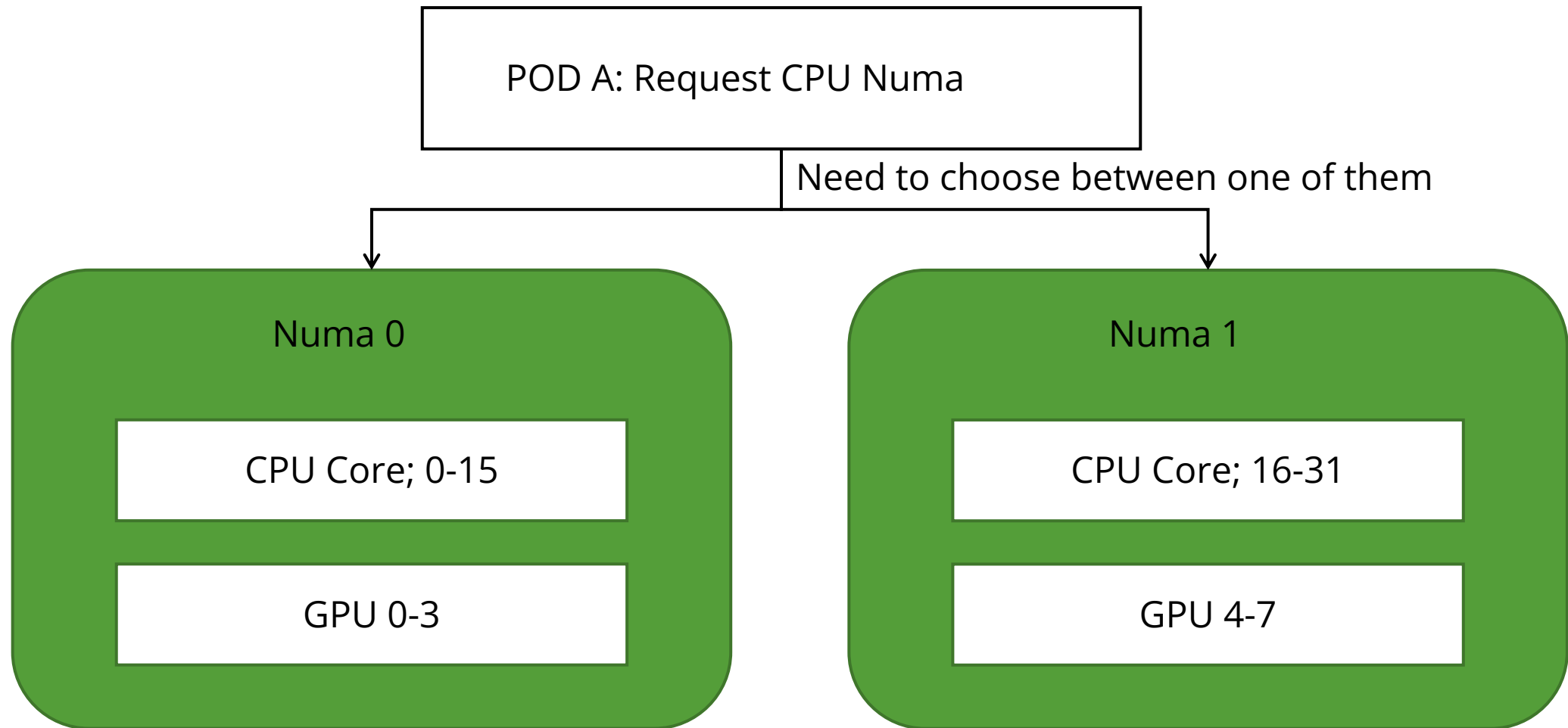
```
>kubectrl port-forward pod/lupine-sessiion 14833:14833
```

```
>export LUPINE_SERVER=127.0.0.1:14833
```

```
>LD_LIBRARY_PATH=/opt/lupine/lib nvidia-smi
```



CPU Numa Align



THANK YOU